



GEMeX: A Large-Scale, Groundable, and Explainable Medical VQA Benchmark for Chest X-ray Diagnosis

Bo Liu¹, Ke Zou^{2,3}, Li-Ming Zhan⁴, Zexin Lu¹, Xiaoyu Dong¹, Yidi Chen⁵, Chengqiang Xie¹,
 Jiannong Cao¹, Xiao-Ming Wu^{1*}, Huazhu Fu^{6*}

¹The Hong Kong Polytechnic University, Hong Kong S.A.R., China, ²National University of Singapore, Singapore

³Sichuan University, China, ⁴Ningbo University of Technology, China, ⁵West China Hospital of Sichuan University, China

⁶Institute of High Performance Computing, Agency for Science, Technology and Research, Singapore

Abstract

Medical Visual Question Answering (Med-VQA) combines computer vision and natural language processing to automatically answer clinical inquiries about medical images. However, current Med-VQA datasets exhibit two significant limitations: (1) they often lack visual and textual explanations for answers, hindering comprehension for patients and junior doctors; (2) they typically offer a narrow range of question formats, inadequately reflecting the diverse requirements in practical scenarios. These limitations pose significant challenges to the development of a reliable and user-friendly Med-VQA system. To address these challenges, we introduce a large-scale, Groundable, and Explainable Medical VQA benchmark for chest X-ray diagnosis (GEMeX), featuring several innovative components: (1) a multi-modal explainability mechanism that offers detailed visual and textual explanations for each question-answer pair; thereby enhancing answer comprehensibility; (2) four question types—open-ended, closed-ended, single-choice, and multiple-choice—to better reflect practical needs. With 151,025 images and 1,605,575 questions, GEMeX is the currently largest chest X-ray VQA dataset. Evaluation of 12 representative large vision language models (LVLMs) on GEMeX reveals suboptimal performance, underscoring the dataset’s complexity. Meanwhile, we propose a strong model by fine-tuning an existing LVLM on the GEMeX training set. The substantial performance improvement showcases the dataset’s effectiveness. The benchmark is available at www.med-vqa.com/GEMeX.

1. Introduction

Large vision language models (LVLMs) have recently made huge advancements in artificial intelligence [1, 2, 13, 34, 38,

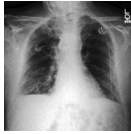
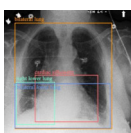
Dataset	VQA Type	Example
 MIMIC-CXR-VQA	Closed-ended	Closed-ended VQA: Q: Is any devices present within the right atrium? A: No
	Open-ended	Open-ended VQA: Q: What are all the diseases identifiable within the right hilar structures? A: A small right pleural effusion.
 Ours	Single-choice	Multi-choice VQA: Q: Which regions on the X-ray show signs of abnormalities? C: ["A: Bilateral lungs", "B: Right costophrenic angle", "C: Bilateral lower lung", "D: Cardiac region"] A: ["A", "B", "C", "D"] Reason: Abnormalities are seen in the bilateral lungs (hyperinflation), right costophrenic angle (blunting), bilateral lower lung (atelectasis), and cardiac silhouette (enlargement).
	Multi-choice	Bounding Box: [[28, 40, 167, 190], [28, 124, 85, 184], [28, 126, 165, 189], [57, 114, 147, 180]]

Figure 1. Our GEMeX stands out from existing medical VQA datasets by providing diverse question types and comprehensive multimodal explanations: textual reasoning and visual grounding.

39, 53, 56], demonstrating remarkable capabilities in understanding visual content while generating coherent natural language responses. These advancements have driven innovations across various domains [15, 47], with healthcare emerging as an important application. Within this domain, medical visual question answering (Med-VQA) stands out as a crucial task that automatically provides reliable and user-friendly answers [29] to questions about medical images [24], facilitating healthcare professionals in diagnosis, medical education, and clinical decision-making.

To ensure the reliability and user-friendliness of Med-VQA systems, it is crucial to incorporate answer explanations along with a diverse set of question formats. Although significant progress has been made by existing Med-VQA systems [16, 18, 24, 30, 51], none have yet integrated explanations for the answers, especially in terms of the visual aspect. As emphasized by [27], explanations are as essential as the answers themselves in general VQA systems. This holds even stronger in medical VQA, where

* Corresponding author.

Dataset	# Images	# QA Pairs	# Modalities	# Question Types‡	# Groundable	# Explainable
VQA-RAD [24]	0.315K	3.5K	Diverse†	O. & C.	✗	✗
SLAKE [30]	0.642K	14K	Diverse†	O. & C.	✗	✗
OmniMedVQA [18]	118.010K	128K	Diverse†	O. & C. & S.	✗	✗
PMC-VQA [51]	149.075K	227K	Diverse†	O. & C. & S.	✗	✗
VQA-Med [7]	4.5K	4.5K	Diverse†	O. & C.	✗	✗
PathVQA [16]	149K	33K	Pathology	O. & C.	✗	✗
RadGenome-Chest CT [52]	50.188K	1.3M	Chest CT	O. & C.	✓	✗
MIMIC-Diff-VQA [17]	164.324K	700K	Chest X-ray	O. & C.	✗	✗
MIMIC-CXR-VQA [4]	142.797K	377K	Chest X-ray	O. & C.	✗	✗
GEMeX (Ours)	151.025K	1.6M	Chest X-ray	O. & C. & S. & M.	✓	✓ (Vision & Language)

Table 1. Comparison of medical VQA Datasets. † indicates a composition of multiple body parts (e.g., head, chest, abdomen) and various imaging modalities (e.g., CT, MRI, X-ray, pathology). In the # Question Types‡ column, O., C., S., and M. represent “Open-ended”, “Closed-ended”, “Single-choice”, and “Multi-choice”, respectively.

the domain-specific nature of the task amplifies the need for clarity [24, 54]. Additionally, the limited range of question formats, such as the absence of multiple-choice questions, restricts the real-world applicability of medical AI systems.

To tackle the aforementioned limitations, we develop a large-scale, **Groundable**, and **Explainable Medical VQA** benchmark for chest **X-ray** diagnosis (GEMeX). We first undertake a comprehensive data refinement process upon the Chest ImaGenome [45]. By collaborating with radiologists, we systematically redefine anatomical regions and establish more precise vision-text correspondence mappings, resulting in accurate region-grounded reports for each X-ray image. Subsequently, we leverage GPT-4o [1] to generate a diverse set of questions based on these grounded reports, covering four categories of varying difficulty levels: open-ended, closed-ended, single-choice, and multiple-choice questions. Each question-answer pair is enriched with explicit reasoning and corresponding visual region annotations, as shown in Figure 1. The resulting dataset comprises 151,025 images and 1,605,575 questions. *Currently, this is the largest VQA dataset for chest X-rays and the first medical VQA dataset that simultaneously includes both textual and visual explanations.*

We evaluate 12 representative LVLMS, including 7 from the general domain (e.g., LLaVA [34], DeepSeek-VL [35], GPT-4o-mini [1]), and 5 from the medical domain (e.g., LLaVA-Med [26], XrayGPT [40], RadFM [43]). The experimental findings underscore the challenging characteristics of our dataset. Additionally, we propose a simple instruction-tuning strategy that derives a task-specific LVLMS. The impressive performance improvement highlights the effectiveness of our dataset. Overall, we develop three metrics for measuring the accuracy of model outputs in terms of answers, reasoning, and visual grounding (localization generation). Notably, we apply both semantics-level score and gram-based metrics of natural language generation (e.g., BLEU and ROUGE) for textual parts. Re-

sults indicate that for models without GEMeX fine-tuning, semantics-level scoring is more reliable. After fine-tuning, however, the natural language generation metrics can better reflect the model’s understanding of the dataset.

This manuscript makes the following key contributions:

- We present GEMeX, a large-scale medical VQA dataset for chest X-rays, designed to support diverse question types and provide enhanced explainability for medical VQA systems. To our knowledge, it is the largest chest X-ray VQA dataset and the first Med-VQA dataset to embody the concept of multimodal explainability.
- We systematically benchmark 12 representative LVLMS using GEMeX, introducing multiple evaluation metrics to comprehensively demonstrate the performance of current popular LVLMS on the Med-VQA task.
- Our method shows that integrating precise vision-text explainability enhances the visual reasoning of LVLMS, addressing a key limitation in many models. We emphasize the need for a large-scale, groundable, and explainable VQA benchmark to advance LVLMS development and deployment in healthcare.

2. Related Work

2.1. Medical VQA Datasets

In recent years, various datasets have been created to advance medical VQA research, each tackling specific challenges in clinical domains. A detailed comparison to other VQA datasets is in Table 1. Specifically, VQA-RAD [24] is a pioneer dataset that offers over 3,000 question-answer pairs focused on radiology images. SLAKE [30] is the first manually created dataset with over 14,000 QA pairs across CT, MRI, and X-ray images, enabling models to handle complex scenarios by combining visual and textual information. VQA-Med [7] is a key dataset for Med-VQA competitions, with 4,500 radiology images and paired QAs. OmniMedVQA [18] provides more data and imag-

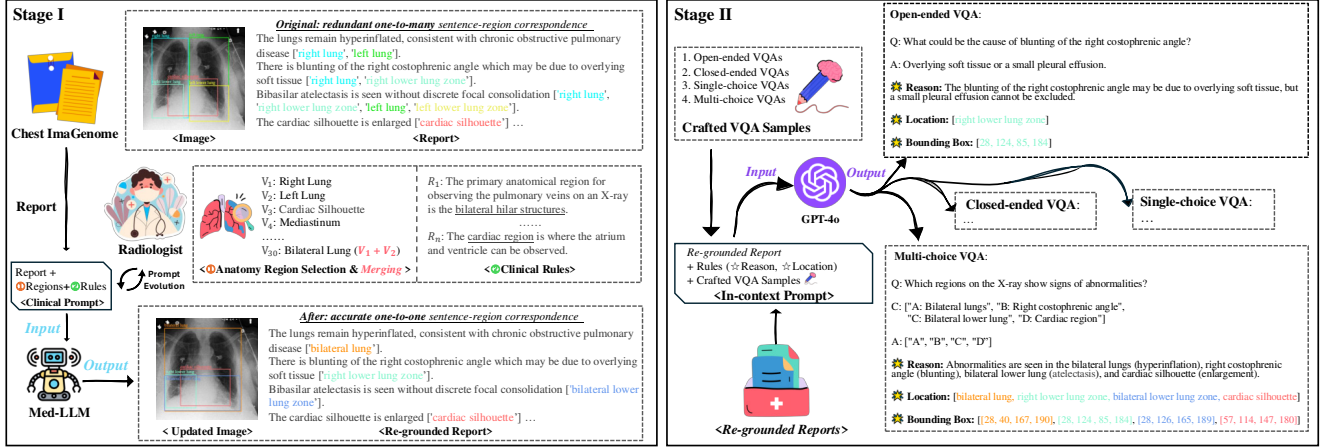


Figure 2. Illustration of the proposed pipeline for constructing our GEMeX, with two main stages. In Stage I (left), medical LLM performs re-grounding on the original reports based on the pathological regions and clinical guidance specified by the radiologists, generating more precise sentence-region correspondence. In Stage II (right), the well-crafted prompt enables GPT-4o to generate a high-quality, large-scale Med-VQA dataset with both textual and visual explanations, leveraging the re-grounded reports from Stage I.

ing modalities, which cover the entire body, to encourage model generalization. PMC-VQA [51] generates VQA data by prompting a large language model to decompose captions of biomedical figures, enabling academic knowledge extraction. PathVQA [16] supplies over 32,000 QA pairs on histopathological images for fine-grained analysis.

For specialized tasks, RadGenome-Chest CT [52] supports chest CT diagnostics, while MIMIC-Diff-VQA [17] emphasizes differential diagnosis reasoning between two X-rays. MIMIC-CXR-VQA [4] expands MIMIC-CXR [21] with diverse question templates to generate thoracic radiology QA pairs, aiding in chest abnormality detection. However, all current datasets lack explainability and diverse question formats. They do not provide detailed visual and textual explanations for answers, which limits their usability for patient and junior doctor comprehension. Additionally, limited question types restrict their ability to simulate the variety of inquiries encountered in practice.

2.2. Medical VQA Methods

Inspired by the advancements in general VQA, medical VQA has gained significant traction as a specialized domain. Due to data limitations, however, most approaches [14, 22, 24, 31, 36, 46] have focused on directly embedding visual and textual information jointly to capture their relationships. With the rise of contrastive language-image pertaining (CLIP) [37], methods [10, 11, 49] start to focus on applying CLIP to Med-VQA. A promising way is to fine-tune CLIP’s joint embeddings to better handle specific medical domains, enhancing the model’s understanding of clinical questions and visual features [28]. Recently, the explosion of large vision language models has further pushed the boundaries of medical domain [3, 25, 26, 40, 43,

57]. Generally, they first pre-train models on a large-scale image-text dataset (like PMC-OA [28], PMC-15M [49]) to map visual features into language model’s embedding space and then further tune with instruction data for medical consultation [3, 34] or disease diagnosis [9, 40, 42, 55]. These models are now leveraged for Med-VQA tasks to provide richer, more context-aware answers, extending beyond simple text-image alignment to incorporate broader knowledge-based reasoning. However, current methods lack explainability, especially in the visual aspect, due to the corresponding absence in current Med-VQA datasets.

3. Construction of GEMeX

We propose a two-stage pipeline for constructing our GEMeX dataset, accompanied by an overview in Figure 2.

3.1. Re-grounding Reports

As shown in stage I of Figure 2, we build upon Chest ImaGenome [45] to construct our dataset, but with an emphasis on the mapping precision between visual regions and textually described entities. After consulting radiologists, we find that the anatomical descriptions of regions of interest in ImaGenome are imprecise and redundant, introducing ambiguity into clinical diagnoses. Specifically, a single sentence can be associated with multiple anatomical regions, e.g., “There is blunting of the right costophrenic angle which may be due to overlying soft tissue” corresponding to “right lung” and “right lower lung zone”. This redundancy poses challenges in training models to precisely visual grounding. Hence, we perform re-grounding to ensure each sentence is linked to a single, precise region.

Anatomical Region Selection and Merging. In the

original Chest ImaGenome, there are 29 significant pathological regions (with bounding boxes). However, in alignment with radiologists’ practices, our dataset focuses on retaining core regions that are crucial for diagnosing diseases through X-rays, such as the “left lower lung” and “mediastinum”. Less significant or marginal areas are excluded to streamline the diagnostic training process and enhance clinical relevance, like “carina” which is not considered a core region, and “clavicle” which accounts for only about 2% of the total regional frequency. Furthermore, to enhance clarity and ensure that each sentence corresponds to a single pathological region with finer granularity, semantically similar regions are merged. For instance, the “left lower lung zone” and “right lower lung zone” are combined into a “bilateral lower lung zone”. This aligns with conditions like “bibasilar atelectasis”, as illustrated in Stage I of Figure 2, where the condition is described as “Bibasilar atelectasis is seen without discrete focal consolidation”. In total, we define 30 anatomical regions. Detailed transformation from Chest ImaGenome to ours can be found in the Appendix.

Report Re-grounding Using Medical LLM. Then, we utilize OpenBioLLM-70B¹, known for its outstanding performance across various medical NLP tasks, to *re-ground reports by re-selecting a region for each sentence*. To test the effectiveness of the prompt, we begin by randomly selecting 100 pairs from the Chest ImaGenome test set, which includes approximately 367 sentences. Initially, the performance of the LLM is suboptimal due to: (1) inner knowledge about X-ray disease observation areas is not sufficiently precise, as OpenBioLLM is an NLP model that lacks clinical expertise, and (2) when a sentence indeed involves multiple regions that cannot be merged, the model may either output multiple regions or arbitrarily select one.

Iterative Prompt Refinement via Radiologist Feedback: To address these limitations, we employ an iterative approach, gradually incorporating clinical guidance from radiologists and manually-labeled such cases (in-context learning) to refine the prompt, facilitating (1) more accurate region selection; and (2) more effective sentence splitting and rewriting. For example, “The cardiomedastinal silhouette is normal.” is converted into {“The cardiac silhouette is normal.”:“cardiac silhouette”, “The mediastinal silhouette is normal.”:“mediastinum”}, where “cardiomedastinal” corresponds to the “cardiac silhouette” and “mediastinum”. This approach ensures the output clauses align one-to-one with the respective regions. Ultimately, the final prompt is determined with an accuracy of approximately 98.4% on the aforementioned test set, provided in the Appendix.

Figure 2 presents an example of a re-grounded report and its corresponding updated image.

¹<https://huggingface.co/aaditya/Llama3-OpenBioLLM-70B>

3.2. Groundable and Explainable VQA Generation

Although there are many Med-VQA datasets [4, 17] available, some even generated using MIMIC-CXR or Chest ImaGenome, they all have two weaknesses that diminish their practicality: (1) lacking strong explainability, especially the visual guidance, that hinders the user’s understanding; (2) a restricted range of question types, typically confined to open-ended or closed-ended formats with no inclusion of choice-based questions, reducing the flexibility and comprehensiveness. In general, these issues highlight the necessity for more versatile and explainable Med-VQA datasets to enhance their utility in clinical settings.

Data Generation with Quality Control. As shown in Stage II of Figure 2, we generate our VQA dataset based on re-grounded reports. Here, we employ GPT-4o (2024-08-06) [1] as a generator due to its remarkable capabilities in understanding and generating long texts. *We ensure the quality of the generated dataset by:* (1) to ensure a diverse range of question content [30], like “abnormality” and “location”, we identified 7 distinct categories through discussions with radiologists, as illustrated in Figure 4. Then, we manually craft questions covering these 7 types for 30 images, which serve as good demonstrations in the prompt to enhance the generation accuracy and better align with our objectives; (2) we also design specific rules (like not generating questions that need to be answered by comparing two images) to ensure the generated VQAs are answerable; (3) similar to the re-grounding process, we extract 50 samples to observe the effect of the prompt. Only when the overall performance meets our expectations do we proceed with the final dataset generation.

For each image-report sample, we instruct the GPT-4o to generate a total of 11 questions: 3 open-ended VQAs, 2 closed-ended VQAs, 3 single-choice VQAs, and 3 multi-choice VQAs², culminating in approximately 1.6 million VQA pairs. The reports containing less than three sentences will not be used to generate QAs. The generated location is provided as an anatomical region in text format (e.g., “left lower lung zone”). To enable the VQA model to identify the specific location on the image, a post-processing step is required to map the region to bounding box coordinates. Figure 2 illustrates multi-modal explainability, with more examples and the final prompt provided in the Appendix.

4. Statistics of GEMeX

Dataset Split. GEMeX is partitioned in accordance with the distribution of MIMIC-CXR. Specifically, we have 149,535 images with 1,588,975 QA pairs for training, 1,190 images with 12,640 QA pairs for validation, and 300 images with 3,960 QA pairs for testing. Detailed statis-

²Single-choice refers to VQA questions with only one correct answer, while multi-choice refers to those with multiple correct answers.

	Open. (T/B)	Closed. (T/B)	Single. (T/B)	Multi. (T/B)
Train	441,471/466,725	272,323/277,249	441,114/448,810	434,067/861,635
Valid	3,524/3,704	2,145/2,184	3,520/3,599	3,451/6,955
Test	1,144/1,189	543/552	1,300/1,310	973/1,870
Total	446,139/471,618	275,011/279,985	445,934/453,719	438,491/870,460

Table 2. Distribution statistics of question types (T) and number of bounding boxes (B) across data splits.

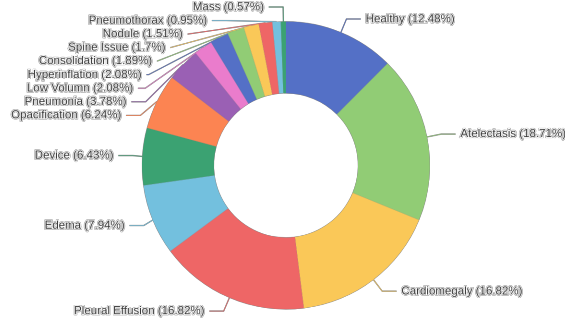


Figure 3. The distribution of normality and common abnormality contained in images from the test set of our GEMeX.

tics, including question type distribution and the number of bounding boxes, are shown in Table 2.

Data Quality. To establish a golden test set, we selected 300 images from the MIMIC-CXR’s test set, initially accompanied by 3,291 questions auto-generated by GPT-4o. Radiologists meticulously reviewed these questions, correcting around 10 incorrect answers and adjusting 3 inaccurate location annotations. This minimal revision rate demonstrates the high quality of the generated dataset. Additionally, the radiologists contributed approximately 600 new questions, thereby creating a comprehensive golden test set for benchmarking large vision language models.

Distribution of Clinical Observations. The X-ray image selection for the test set is according to the clinical observations. In Figure 3, we plot the distribution of normality and abnormality contained in images. The original distribution of MIMIC-CXR, characterized by a high proportion of healthy samples, introduces a significant bias affecting model performance [32]. To mitigate this issue, we preserve only a small proportion of healthy samples (around 12.48%) during manual cleaning. Meanwhile, we ensure that clinically important observations occupy a large proportion, such as “atelectasis”, “cardiomegaly”, “edema”, and “pleural effusion” [19]. Additionally, common diseases or observations like “pneumonia”, “opacification”, and “pneumothorax” are also included.

Distribution of Question Content. We show the distribution of the question content categories that GPT-4o itself summarizes according to demonstrations during VQA generation. Figure 4 shows the corresponding results

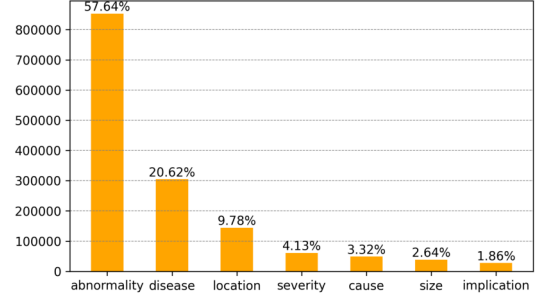


Figure 4. The distribution of question content in our GEMeX.

where we can find that “abnormality”, “disease”, and “location” account for over 88%, while the remaining categories mainly include “cause”, “size”, “severity”, and “implication”, which highlights the diversity of questions.

5. Evaluation

5.1. Models

A Strong Baseline Fine-Tuned on GEMeX. To validate the effectiveness of the dataset, especially the auto-generated training set, we propose a question-type-aware instruction tuning to fine-tune LLaVA-Med-v1-7B [26] on the training set of GEMeX, termed as LLaVA-Med-GEMeX, serving as a strong baseline. Specifically, for each VQA sample from our GEMeX, we add a type prompt $\mathbf{X}_{\text{Type}}$ after the original system prompt and a question $\mathbf{X}_{\text{Question}}$ with its answer $\mathbf{X}_{\text{Answer}}$, textual reason $\mathbf{X}_{\text{Reason}}$, and corresponding visual location $\mathbf{X}_{\text{Location}}$, constructing a single-turn dialogue as in Table 3. Generally, $\mathbf{X}_{\text{Type}}$ is “Input a {Type} question, and the assistant will output its answer {Supplement} with a detailed reason and corresponding visual location.” where {Type} refers to “open-ended/closed-ended/single-choice/multi-choice” and {Supplement} is replaced by “none/(yes or no)/(an option)/(some options)”, respectively. Some input samples, as well as the fine-tuning details, are shown in the Appendix.

$\langle \text{Ori_System_Prompt} \rangle \mathbf{X}_{\text{Type}} \langle \text{STOP} \rangle$ Human: $\langle \text{image} \rangle \backslash \mathbf{X}_{\text{Question}} \mathbf{X}_{\text{Choices}} \text{ (if any)}$ $\langle \text{STOP} \rangle$ Assistant: $\langle \text{answer} \rangle \mathbf{X}_{\text{Answer}} \langle \text{reason} \rangle \mathbf{X}_{\text{Reason}}$ $\langle \text{location} \rangle \mathbf{X}_{\text{Location}} \langle \text{STOP} \rangle$
--

Table 3. Input format for fine-tuning LLaVA-Med.

Existing LVLMS. Besides fine-tuning a task-oriented model, we perform a zero-shot evaluation on our GEMeX dataset across the other 12 LVLMS, with 7 in the general domain and the other 5 in the medical domain:

- **In the General Domain:** we test LLaVA-v1 [34], Mini-GPT4-v1 [56], mPLUG-Owl [48],

Models	Open-ended		Closed-ended			Single-choice			Multi-choice			Avg.†
	AR-score†	V-score	A-score	AR-score†	V-score	A-score	AR-score†	V-score	A-score	AR-score†	V-score	
Random	-	-	48.80	-	-	25.85	-	-	7.50	-	-	-
GPT-4o-mini [1]	97.68	<u>18.05</u>	59.30	71.14	<u>28.64</u>	<u>59.00</u>	<u>77.47</u>	<u>23.62</u>	<u>49.13</u>	<u>82.91</u>	<u>19.19</u>	<u>82.30</u>
LLaVA-v1 [34]	76.14	-	30.76	38.02	-	-	50.47	-	-	66.52	-	57.79
LLaVA-v1.5 [33]	77.62	-	58.93	57.00	-	47.00	57.05	-	-	65.17	-	64.21
Mini-GPT4-v1 [56]	55.32	-	26.33	31.09	-	-	37.63	-	-	46.65	-	42.67
mPLUG-Owl [48]	76.73	-	27.26	36.70	-	32.00	46.89	-	-	67.92	-	57.06
DeepSeek-VL [35]	79.30	11.00	57.10	59.86	8.28	51.69	62.03	8.57	17.99	70.35	12.98	67.89
Qwen-VL-Chat [6]	78.36	3.17	23.02	45.79	12.25	44.69	59.15	16.69	7.30	67.21	2.26	62.63
LLaVA-Med-v1 [26]	90.34	-	62.62	69.91	-	-	61.74	-	-	68.14	-	72.53
LLaVA-Med-v1.5 [26]	94.43	-	<u>71.82</u>	<u>76.54</u>	-	-	66.04	-	-	67.28	-	76.07
MiniGPT-Med [3]	86.12	-	55.24	65.25	-	-	55.61	-	-	64.33	-	67.83
XrayGPT [40]	81.17	-	-	68.17	-	-	48.33	-	-	55.10	-	63.19
RadFM [43]	88.57	-	58.01	67.91	-	-	57.82	-	-	62.41	-	69.18
LLaVA-Med-GEMeX	<u>97.05</u>	51.47	77.35	80.72	53.20	73.08	81.42	54.57	67.42	84.98	47.99	86.04

Table 4. Performance of representative LVLMs on our GEMeX across four different question types. The AR-score combines answer and reasoning to evaluate textual output performance. † denotes the GPTScore value (%). The A-score indicates answer or choice accuracy (%), and the V-score represents mIoU (%). The best results are bolded, and the second-best are underlined in each column.

LLaVA-v1.5 [33], DeepSeek-VL [35], Qwen-VL-Chat [6], and GPT-4o-mini (2024-07-18). We did not test GPT-4o due to its safety policy prohibiting medical image analysis, especially when asking about the condition of patients.

- **In the Medical Domain:** we evaluate LLaVA-Med-v1 [26], LLaVA-Med-v1.5 [26], MiniGPT-Med [3], XrayGPT [40], and RadFM [43].

5.2. Evaluation Metrics

In GEMeX, each question has a corresponding answer, textual reason, and visual location. Ideally, we aim to evaluate all these three aspects with designed metrics as follows:

- **Answer-Reason Score (AR-score):** In reality, most LVLMs struggle to generate accurate outputs in terms of format. This doesn’t necessarily mean these models lack the knowledge to answer the questions but rather simply lack the ability to follow instructions properly. To ensure a fair comparison, we introduce the Answer-Reason score (AR-score) as an evaluation metric for the textual output, where the answer and reason parts from each test sample are merged as a reference (ground truth), and the evaluated LVLm’s output serve as a candidate. We use GPTScore [26] to calculate the AR-Score from a semantic perspective. Specifically, GPT-4o is leveraged to quantify the correctness by treating the aforementioned reference as a textual response from assistant #1, while the candidate as the response from assistant #2. With both responses, the original question, and the X-ray report, GPT-4o assesses the accuracy, relevance, and helpfulness of each assistant’s answer and provides an overall score on a scale of 1 to 10, where a higher score indicates better performance. We then calculate the relative

score using GPT-4o’s reference score for normalization. Besides, we also employ common NLG metrics (*e.g.*, BERTScore [50], BLEU, ROUGE) to evaluate AR-score. The detailed prompt can be found in the Appendix.

- **Answer Score (A-score):** For responses where the model can output specific answers (such as yes/no for closed-ended questions or options for single/multiple choice questions), we calculate the accuracy by comparing with the ground truth. It is worth noting that although some models cannot directly output the answer, we still attempt to match it from their responses.
- **Visual Score (V-score):** For models capable of visual grounding (*i.e.*, outputting visual locations), we calculated mean intersection over union (mIoU) as a measurement. For a VQA case, considering there might be multiple corresponding locations (commonly seen in multiple-choice questions), we use the Hungarian algorithm [23] to match the predicted bounding boxes with the actual ones.

5.3. Results and Analysis

Overall Performance. The comprehensive results are shown in Table 4. The first 7 rows indicate the performance of general LVLMs, while the last 6 rows present the results of medical ones and our fine-tuned version of LLaVA-Med-v1 (termed as LLaVA-Med-GEMeX). It can be found that:

- **Most existing LVLMs exhibit weak performance when tested on GEMeX.** The only exception is GPT-4o-mini, which achieves an AR-score above 80 on average across all tasks. When considering specific question types, LLaVA-Med (both versions 1 and 1.5) stands out for its strong performance on open-ended questions, scoring above 90 on the AR-score. However, all models show poor results on the other three categories of tasks.

Models	Open-ended			Closed-ended			Single-choice			Multi-choice		
	BERTScore	ROUGE-L	BLEU-1	BERTScore	ROUGE-L	BLEU-1	BERTScore	ROUGE-L	BLEU-1	BERTScore	ROUGE-L	BLEU-1
GPT-4o-mini [1]	30.43	22.67	18.25	40.02	25.63	19.10	<u>48.34</u>	<u>39.17</u>	<u>30.82</u>	<u>46.58</u>	<u>39.20</u>	<u>28.65</u>
LLaVA-v1 [34]	20.09	15.22	11.57	22.42	13.10	8.01	20.25	14.97	10.61	19.69	17.35	11.15
LLaVA-v1.5 [33]	21.49	16.11	12.20	32.59	15.37	6.69	17.42	17.53	1.49	23.74	21.20	8.95
Mini-GPT4-v1 [56]	15.03	14.66	11.46	13.83	9.65	6.31	6.50	6.79	4.60	5.31	5.79	3.22
mPLUG-Owl [48]	22.52	17.03	13.22	32.23	20.20	13.92	39.64	33.69	30.32	26.09	24.97	16.68
DeepSeek-VL [35]	24.06	18.62	15.94	26.12	23.27	13.83	26.16	30.46	18.79	22.10	27.32	20.74
Qwen-VL-Chat [6]	23.31	18.48	14.63	33.18	22.43	17.19	25.47	22.25	6.95	22.03	22.88	12.26
LLaVA-Med-v1 [26]	25.14	19.63	15.93	38.04	29.08	19.74	34.89	30.10	25.84	28.63	26.51	20.99
LLaVA-Med-v1.5 [26]	26.42	21.38	17.28	<u>44.48</u>	<u>36.73</u>	<u>26.35</u>	36.62	30.32	25.44	28.11	24.49	16.53
MiniGPT-Med [3]	23.47	19.20	16.03	34.31	29.47	19.13	30.11	28.51	22.13	26.51	24.42	15.98
XrayGPT [40]	22.57	18.30	15.73	21.35	14.55	10.17	16.31	12.17	9.23	12.15	10.30	6.22
RadFM [43]	24.96	20.71	17.73	37.43	27.95	20.56	32.30	27.02	24.39	25.81	20.02	13.80
LLaVA-Med-GEMeX	42.69	32.75	25.28	54.44	38.39	33.99	56.35	53.23	47.31	54.95	50.85	43.99

Table 5. Performance of representative LVLs evaluated using various natural language generation metrics for AR-score, including BERTScore, ROUGE-L, and BLEU-1. The best results are bolded, and the second-best are underlined in each column.

- **When faced with choice-based questions, most models, particularly those in the medical domain, struggle to provide definitive answers, despite their ability to analyze each option.** This accounts for why many models have an AR-score but lack an A-score, highlighting the importance of introducing these types of questions.
- **Powerful LVLs, such as GPT-4o-mini, often rely on shortcut reasoning rather than real multimodal reasoning.** Although these models can sometimes answer questions to a certain extent (as indicated by the AR-scores), they often fail to accurately achieve visual grounding. This suggests that these models tend to address Med-VQA tasks using shortcut knowledge, such as retrieving information from their pre-training memory, instead of engaging in real multimodal reasoning [8]. However, real multimodal reasoning is essential for the explainability of Med-VQA systems.
- **Through simple question-type-aware instruction tuning, the proposed baseline model achieves a significant performance improvement,** with an approximate **13.5%** increase in average AR-score compared to LLaVA-Med-v1. Notably, it surpasses GPT-4o-mini on most metrics, demonstrating the reliability of the training set. However, a substantial gap remains for practical application, highlighting the challenges associated with GEMeX.

Limitation. Note that the proposed baseline model is inherently task-specific, which may result in reduced accuracy on other tasks or a diminished capacity for conversational engagement. The true potential of our GEMeX lies in its integration into multi-task training frameworks, such as the second training stage of LLaVA-Med. The baseline model primarily serves to demonstrate the dataset’s effectiveness while also providing a robust benchmark.

More Metrics. As mentioned in Section 5.2, we also calculate NLG metrics to measure AR-score. Detailed results

are shown in Table 5. Overall, the NLG metrics generally share the same trend as GPTScore (AR-score in Table 4), but there are some minor differences. (1) **High NLG scores do not always correlate with strong model performance,** as seen with mPLUG-Owl compared to LLaVA-v1.5. Essentially, LLaVA-1.5 demonstrates higher performance, such as achieving an answer accuracy rate (A-score) in single-choice tasks that is **15%** higher than that of mPLUG-Owl. However, since LLaVA-v1.5’s output mostly consists of the answer without reason, the shorter output results in a lower NLG score, with its BLEU-1 approximately **28.8%** lower than mPLUG-Owl; (2) **Fine-tuning on GEMeX results in more pronounced improvements in NLG metrics compared to GPTScore.** For example, the fine-tuned baseline model shows only a 3.7% average improvement over GPT-4o-mini on GPTScore, but achieves a 12.1% improvement on average NLG metrics. This significant enhancement better reflects the model’s learning from the dataset. We suggest using GPTScore for models not fine-tuned on GEMeX due to its focus on semantic understanding, while NLG metrics are preferred for fine-tuned models as they better capture alignment with the dataset.

Transferability. To further validate the effectiveness of GEMeX, we assess it from a transfer learning perspective by treating GEMeX as a pre-training dataset and subsequently testing our LLaVA-Med-GEMeX model on other X-ray VQA datasets. Considering potential overlap in data sources, we utilize the CXR portion of the SLAKE test set [30], which includes 505 open-ended questions and 235 closed-ended questions. The zero-shot performance of LLaVA-Med-GEMeX and LLaVA-Med-v1 on this dataset is reported in Table 7. The results indicate that our LLaVA-Med-GEMeX significantly outperforms LLaVA-Med-v1 in both tasks, demonstrating the effective X-ray knowledge acquired from GEMeX.

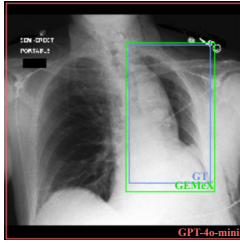
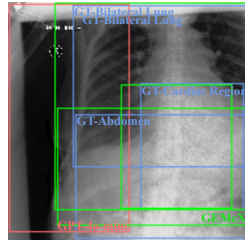
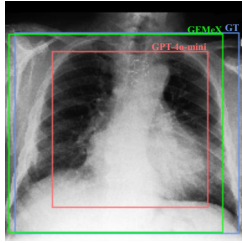
Challenging examples from GEMeX:			
	CASE I  (Bbox predicted by each model)	CASE II  (Bbox predicted by each model)	CASE III  (Bbox predicted by each model)
Question	What might be the cause of the overexpansion of the right lung?	What regions on the X-ray indicate abnormal findings? Choices: [A: Bilateral lung, B: Cardiac region, C: Abdomen, D: Spine]	What is the overall condition of the lungs based on the CXR? Choices: [A: Clear, B: Consolidated, C: Effused, D: Thickened]
Ground Truth	<Answer>: Compensatory hyperinflation due to left lung removal. <Reason>: After a pneumonectomy, the remaining lung may overexpand to compensate for the loss of the left lung .	<Answer>: [A, B, C]. <Reason>: The bilateral lung shows pleural effusion , the cardiac region shows enlargement, and the abdomen shows the nasogastric tube . The spine is normal.	<Answer>: A. <Reason>: The CXR indicates that the lungs are clear of consolidation or effusion , although right pleural-based thickening at the base.
GPT-4o-mini	<Answer>: The overexpansion of the right lung may be caused by conditions such as a right-sided pneumothorax or chronic obstructive pulmonary disease (COPD). <Reason>: Overexpansion can occur when air enters the pleural space (as in pneumothorax) or when there is an obstruction that prevents air from leaving the lung, causing it to become hyperinflated. In COPD, the lung tissue loses elasticity, leading to overinflation. <GPTScore>: 44.44	<Answer>: [A: Bilateral lung]. <Reason>: The bilateral lung region shows potential abnormalities such as opacities or other irregularities that may indicate pathology (✗) . <GPTScore>: 37.50	<Answer>: B. <Reason>: The presence of increased opacity in the lung fields suggests consolidation , indicating that the lungs are not clear and may have fluid or solid material in the alveoli (✗). <GPTScore>: 33.33.
LLaVA-Med-GEMeX	<Answer>: Possible chronic lung disease or compensatory mechanism (✓). <Reason>: Overexpansion of the lung can be due to chronic lung diseases or compensatory mechanisms, especially after a left lobectomy (✓) . <GPTScore>: 66.67	<Answer>: [A, B, C]. <Reason>: The bilateral lung region shows interstitial opacities (✗) , the cardiac region shows enlargement (✓), and the abdomen shows the feeding tube location (✓) . <GPTScore>: 87.50	<Answer>: A. <Reason>: The CXR indicates that the lungs are otherwise clear except for the noted pleural thickening (✓) . <GPTScore>: 88.89

Table 6. Challenging questions from GEMeX answered by GPT-4o-mini and our LLaVA-Med-GEMeX, **highlighting the limitations of LVLMS like GPT-4o-mini in visual reasoning, which can be enhanced by integrating knowledge from GEMeX**. Colors in outputs highlight pathological descriptions that are similar to the ground truth, though they may be either correct (✓) or incorrect (✗).

Models	Open-ended AR-Score	Closed-ended	
		A-Score	AR-Score
LLaVA-Med-v1	73.31	56.17	62.35
LLaVA-Med-GEMeX	82.78	69.79	75.06

Table 7. Performance of LVLMS on SLAKE-CXR.

Case Study. In Table 6, we show some questions with outputs from both GPT-4o-mini and our fine-tuned LLaVA-Med-GEMeX for qualitative comparison. In CASE I, although GPT-4o-mini can generate a very detailed answer, it provides answers without reasoning on the visual content, resulting in a significant difference from the ground truth. In contrast, the LLaVA-Med-GEMeX offers relatively accurate visual clues and is able to provide partially correct answers (“the compensatory mechanism”), although there is a false mention of “possible chronic lung disease” when considering the patient’s condition. In CASE II, although GPT-4o-mini can analyze images, its limited capabilities result in selecting only one option and providing a vague reason. In contrast, the LLaVA-Med-GEMeX outputs the correct options but gives an incorrect reason for one op-

tion (*i.e.*, answer “A”). In CASE III, GPT-4o-mini cannot both visually reason and output answers correctly, while the fine-tuned model can give better outputs from these two aspects. From these examples, we can conclude that some LVLMS still lack sufficient understanding of medical images. Meanwhile, while the proposed simple fine-tuning method improves performance, it is still far from fully accurate, leaving much room for further exploration.

6. Conclusion

In this paper, we introduce a benchmark, GEMeX, designed to advance the field of medical VQA with two primary advantages: multimodal explainability and diverse question types. GEMeX not only provides more accessible medical explanations for patients and junior doctors but also serves as a valuable training resource for developing next-generation medical LVLMS with enhanced instruction-following capabilities. We demonstrate the effectiveness and difficulty of the dataset through comprehensive testing of representative LVLMS as well as task-specific fine-tuning, hoping that GEMeX can promote medical VQA development and improve AI-assisted medical care.

Acknowledgment

We thank the reviewers for their valuable feedback. This work was supported by HK RGC Theme-based Research Scheme (No. T43-518/24-N); Project P0056021 under Parent Project P0050643 of the Otto Poon Charitable Foundation Smart Cities Research Institute, HK PolyU; the Research Student Attachment Programme, HK PolyU; and the A*STAR Central Research Fund (“Robust and Trustworthy AI system for Multi-modality Healthcare”), Singapore.

References

- [1] Josh Achiam, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, Florencia Leoni Aleman, Diogo Almeida, Janko Altenschmidt, Sam Altman, Shyamal Anadkat, et al. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774*, 2023. 1, 2, 4, 6, 7, 5
- [2] Jean-Baptiste Alayrac, Jeff Donahue, Pauline Luc, Antoine Miech, Iain Barr, Yana Hasson, Karel Lenc, Arthur Mensch, Katherine Millican, Malcolm Reynolds, et al. Flamingo: a visual language model for few-shot learning. *Advances in neural information processing systems*, 35:23716–23736, 2022. 1
- [3] Asma Alkhalidi, Raneem Alnajim, Layan Alabdullatef, Rawan Alyahya, Jun Chen, Deyao Zhu, Ahmed Alsinan, and Mohamed Elhoseiny. Minigpt-med: Large language model as a general interface for radiology diagnosis. *arXiv preprint arXiv:2407.04106*, 2024. 3, 6, 7, 5
- [4] Seongsu Bae, Daeun Kyung, Jaehee Ryu, Eunbyeol Cho, Gyubok Lee, Sunjun Kweon, Jungwoo Oh, Lei Ji, Eric Chang, Tackeun Kim, et al. Ehrxqa: A multi-modal question answering dataset for electronic health records with chest x-ray images. *Advances in Neural Information Processing Systems*, 36, 2024. 2, 3, 4
- [5] Jinze Bai, Shuai Bai, Shusheng Yang, Shijie Wang, Sinan Tan, Peng Wang, Junyang Lin, Chang Zhou, and Jingren Zhou. Qwen-vl: A versatile vision-language model for understanding, localization, text reading, and beyond. *arXiv preprint arXiv:2308.12966*, 1(2):3, 2023. 5
- [6] Jinze Bai, Shuai Bai, Shusheng Yang, Shijie Wang, Sinan Tan, Peng Wang, Junyang Lin, Chang Zhou, and Jingren Zhou. Qwen-vl: A versatile vision-language model for understanding, localization, text reading, and beyond, 2023. 6, 7
- [7] Asma Ben Abacha, Mourad Sarrouiti, Dina Demner-Fushman, Sadid A Hasan, and Henning Müller. Overview of the vqa-med task at imageclef 2021: Visual question answering and generation in the medical domain. In *Proceedings of the CLEF 2021 Conference and Labs of the Evaluation Forum-working notes*. 21-24 September 2021, 2021. 2
- [8] Jiawei Chen, Dingkan Yang, Tong Wu, Yue Jiang, Xiaolu Hou, Mingcheng Li, Shunli Wang, Dongling Xiao, Ke Li, and Lihua Zhang. Detecting and evaluating medical hallucinations in large vision language models. *arXiv preprint arXiv:2406.10185*, 2024. 7
- [9] Xiaolan Chen, Ziwei Zhao, Weiyi Zhang, Pusheng Xu, Le Gao, Mingpu Xu, Yue Wu, Yinwen Li, Danli Shi, and Ming-guang He. Eyegpt: Ophthalmic assistant with large language models. *arXiv preprint arXiv:2403.00840*, 2024. 3
- [10] Zhihong Chen, Yuhao Du, Jinpeng Hu, Yang Liu, Guanbin Li, Xiang Wan, and Tsung-Hui Chang. Multi-modal masked autoencoders for medical vision-and-language pre-training. In *International Conference on Medical Image Computing and Computer-Assisted Intervention*, pages 679–689. Springer, 2022. 3
- [11] Zhihong Chen, Shizhe Diao, Benyou Wang, Guanbin Li, and Xiang Wan. Towards unifying medical vision-and-language pre-training via soft prompts. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 23403–23413, 2023. 3
- [12] Wei-Lin Chiang, Zhuohan Li, Zi Lin, Ying Sheng, Zhanghao Wu, Hao Zhang, Lianmin Zheng, Siyuan Zhuang, Yonghao Zhuang, Joseph E Gonzalez, et al. Vicuna: An open-source chatbot impressing gpt-4 with 90%* chatgpt quality. See <https://vicuna.lmsys.org> (accessed 14 April 2023), 2(3):6, 2023. 5
- [13] Wenliang Dai, Junnan Li, Dongxu Li, Anthony Meng Huat Tiong, Junqi Zhao, Weisheng Wang, Boyang Li, Pascale Fung, and Steven Hoi. Instructblip: Towards general-purpose vision-language models with instruction tuning, 2023. 1
- [14] Haifan Gong, Guanqi Chen, Sishuo Liu, Yizhou Yu, and Guanbin Li. Cross-modal self-attention with multi-task pre-training for medical visual question answering. In *Proceedings of the 2021 international conference on multimedia retrieval*, pages 456–460, 2021. 3
- [15] Zhaopeng Gu, Bingke Zhu, Guibo Zhu, Yingying Chen, Ming Tang, and Jinqiao Wang. Anomalygpt: Detecting industrial anomalies using large vision-language models. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 1932–1940, 2024. 1
- [16] Xuehai He, Yichen Zhang, Luntian Mou, Eric Xing, and Pengtao Xie. Pathvqa: 30000+ questions for medical visual question answering. *arXiv preprint arXiv:2003.10286*, 2020. 1, 2, 3
- [17] Xinyue Hu, Lin Gu, Qiyuan An, Mengliang Zhang, Liangchen Liu, Kazuma Kobayashi, Tatsuya Harada, Ronald M Summers, and Yingying Zhu. Expert knowledge-aware image difference graph representation learning for difference-aware medical visual question answering. In *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, pages 4156–4165, 2023. 2, 3, 4
- [18] Yutao Hu, Tianbin Li, Quanfeng Lu, Wenqi Shao, Junjun He, Yu Qiao, and Ping Luo. Omnimedvqa: A new large-scale comprehensive evaluation benchmark for medical lvlm. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 22170–22183, 2024. 1, 2
- [19] Jeremy Irvin, Pranav Rajpurkar, Michael Ko, Yifan Yu, Silvana Ciurea-Ilcus, Chris Chute, Henrik Marklund, Behzad Haghighi, Robyn Ball, Katie Shpanskaya, et al. Chexpert: A large chest radiograph dataset with uncertainty labels and expert comparison. In *Proceedings of the AAAI conference on artificial intelligence*, pages 590–597, 2019. 5

- [20] Albert Q Jiang, Alexandre Sablayrolles, Arthur Mensch, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Florian Bressand, Gianna Lengyel, Guillaume Lample, Lucile Saulnier, et al. Mistral 7b. *arXiv preprint arXiv:2310.06825*, 2023. 5
- [21] Alistair EW Johnson, Tom J Pollard, Seth J Berkowitz, Nathaniel R Greenbaum, Matthew P Lungren, Chih-ying Deng, Roger G Mark, and Steven Horng. Mimic-cxr, a de-identified publicly available database of chest radiographs with free-text reports. *Scientific data*, 6(1):317, 2019. 3
- [22] Yash Khare, Viraj Bagal, Minesh Mathew, Adithi Devi, U Deva Priyakumar, and CV Jawahar. Mmbert: Multi-modal bert pretraining for improved medical vqa. In *2021 IEEE 18th International Symposium on Biomedical Imaging (ISBI)*, pages 1033–1036. IEEE, 2021. 3
- [23] Harold W Kuhn. The hungarian method for the assignment problem. *Naval research logistics quarterly*, 2(1-2):83–97, 1955. 6
- [24] Jason J Lau, Soumya Gayen, Asma Ben Abacha, and Dina Demner-Fushman. A dataset of clinically generated visual questions and answers about radiology images. *Scientific data*, 5(1):1–10, 2018. 1, 2, 3
- [25] Suhyeon Lee, Won Jun Kim, Jinho Chang, and Jong Chul Ye. Llm-cxr: Instruction-finetuned llm for cxr image understanding and generation. *arXiv preprint arXiv:2305.11490*, 2023. 3
- [26] Chunyuan Li, Cliff Wong, Sheng Zhang, Naoto Usuyama, Haotian Liu, Jianwei Yang, Tristan Naumann, Hoifung Poon, and Jianfeng Gao. Llava-med: Training a large language-and-vision assistant for biomedicine in one day. *Advances in Neural Information Processing Systems*, 36, 2024. 2, 3, 5, 6, 7, 4
- [27] Qing Li, Qingyi Tao, Shafiq Joty, Jianfei Cai, and Jiebo Luo. Vqa-e: Explaining, elaborating, and enhancing your answers for visual questions. In *Proceedings of the European Conference on Computer Vision (ECCV)*, pages 552–567, 2018. 1
- [28] Weixiong Lin, Ziheng Zhao, Xiaoman Zhang, Chaoyi Wu, Ya Zhang, Yanfeng Wang, and Weidi Xie. Pmc-clip: Contrastive language-image pre-training using biomedical documents. In *International Conference on Medical Image Computing and Computer-Assisted Intervention*, pages 525–536. Springer, 2023. 3
- [29] Zhihong Lin, Donghao Zhang, Qingyi Tao, Danli Shi, Gholamreza Haffari, Qi Wu, Mingguang He, and Zongyuan Ge. Medical visual question answering: A survey. *Artificial Intelligence in Medicine*, 143:102611, 2023. 1
- [30] Bo Liu, Li-Ming Zhan, Li Xu, Lin Ma, Yan Yang, and Xiao-Ming Wu. Slake: A semantically-labeled knowledge-enhanced dataset for medical visual question answering. In *2021 IEEE 18th International Symposium on Biomedical Imaging (ISBI)*, pages 1650–1654. IEEE, 2021. 1, 2, 4, 7
- [31] Bo Liu, Li-Ming Zhan, Li Xu, and Xiao-Ming Wu. Medical visual question answering via conditional reasoning and contrastive learning. *IEEE transactions on medical imaging*, 42(5):1532–1545, 2022. 3
- [32] Fenglin Liu, Shen Ge, Yuexian Zou, and Xian Wu. Competence-based multimodal curriculum learning for medical report generation. *arXiv preprint arXiv:2206.14579*, 2022. 5
- [33] Haotian Liu, Chunyuan Li, Yuheng Li, and Yong Jae Lee. Improved baselines with visual instruction tuning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 26296–26306, 2024. 6, 7, 5
- [34] Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. Visual instruction tuning. *Advances in neural information processing systems*, 36, 2024. 1, 2, 3, 5, 6, 7
- [35] Haoyu Lu, Wen Liu, Bo Zhang, Bingxuan Wang, Kai Dong, Bo Liu, Jingxiang Sun, Tongzheng Ren, Zhuoshu Li, Hao Yang, et al. Deepseek-vl: towards real-world vision-language understanding. *arXiv preprint arXiv:2403.05525*, 2024. 2, 6, 7, 5
- [36] Jong Hak Moon, Hyungyung Lee, Woncheol Shin, Young-Hak Kim, and Edward Choi. Multi-modal understanding and generation for medical images and text via vision-language pre-training. *IEEE Journal of Biomedical and Health Informatics*, 26(12):6070–6080, 2022. 3
- [37] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. Learning transferable visual models from natural language supervision. In *International conference on machine learning*, pages 8748–8763. PMLR, 2021. 3
- [38] Aditya Ramesh, Mikhail Pavlov, Gabriel Goh, Scott Gray, Chelsea Voss, Alec Radford, Mark Chen, and Ilya Sutskever. Zero-shot text-to-image generation. In *International conference on machine learning*, pages 8821–8831. Pmlr, 2021. 1
- [39] Andrew Sellergrén, Sahar Kazemzadeh, Tiam Jaroensri, Atilla Kiraly, Madeleine Traverse, Timo Kohlberger, Shawn Xu, Fayaz Jamil, Cían Hughes, Charles Lau, et al. Medgemma technical report. *arXiv preprint arXiv:2507.05201*, 2025. 1
- [40] Omkar Thawkar, Abdelrahman Shaker, Sahal Shaji Mullappilly, Hisham Cholakkal, Rao Muhammad Anwer, Salman Khan, Jorma Laaksonen, and Fahad Shahbaz Khan. Xraygpt: Chest radiographs summarization using medical vision-language models. *arXiv preprint arXiv:2306.07971*, 2023. 2, 3, 6, 7, 5
- [41] Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, Eric Hambro, Faisal Azhar, et al. Llama: Open and efficient foundation language models. *arXiv preprint arXiv:2302.13971*, 2023. 5
- [42] Xiyue Wang, Junhan Zhao, Eliana Marostica, Wei Yuan, Jietian Jin, Jiayu Zhang, Ruijiang Li, Hongping Tang, Kanran Wang, Yu Li, et al. A pathology foundation model for cancer diagnosis and prognosis prediction. *Nature*, pages 1–9, 2024. 3
- [43] Chaoyi Wu, Xiaoman Zhang, Ya Zhang, Yanfeng Wang, and Weidi Xie. Towards generalist foundation model for radiology. *arXiv preprint arXiv:2308.02463*, 2023. 2, 3, 6, 7, 5
- [44] Chaoyi Wu, Weixiong Lin, Xiaoman Zhang, Ya Zhang, Weidi Xie, and Yanfeng Wang. Pmc-llama: toward building open-source language models for medicine. *Journal of the*

- American Medical Informatics Association*, page ocae045, 2024. 5
- [45] Joy T Wu, Nkechinyere N Agu, Ismini Lourentzou, Arjun Sharma, Joseph A Paguio, Jasper S Yao, Edward C Dee, William Mitchell, Satyananda Kashyap, Andrea Giovannini, et al. Chest imagenome dataset for clinical reasoning. *arXiv preprint arXiv:2108.00316*, 2021. 2, 3
 - [46] Li Xu, Bo Liu, Ameer Hamza Khan, Lu Fan, and Xiao-Ming Wu. Multi-modal pre-training for medical vision-language understanding and generation: An empirical study with a new benchmark. *arXiv preprint arXiv:2306.06494*, 2023. 3
 - [47] Peng Xu, Wenqi Shao, Kaipeng Zhang, Peng Gao, Shuo Liu, Meng Lei, Fanqing Meng, Siyuan Huang, Yu Qiao, and Ping Luo. Lvlm-ehub: A comprehensive evaluation benchmark for large vision-language models. *arXiv preprint arXiv:2306.09265*, 2023. 1
 - [48] Qinghao Ye, Haiyang Xu, Guohai Xu, Jiabo Ye, Ming Yan, Yiyang Zhou, Junyang Wang, Anwen Hu, Pengcheng Shi, Yaya Shi, et al. mplug-owl: Modularization empowers large language models with multimodality. *arXiv preprint arXiv:2304.14178*, 2023. 5, 6, 7
 - [49] Sheng Zhang, Yanbo Xu, Naoto Usuyama, Hanwen Xu, Jaspreet Bagga, Robert Tinn, Sam Preston, Rajesh Rao, Mu Wei, Naveen Valluri, et al. Biomedclip: a multimodal biomedical foundation model pretrained from fifteen million scientific image-text pairs. *arXiv preprint arXiv:2303.00915*, 2023. 3
 - [50] Tianyi Zhang, Varsha Kishore, Felix Wu, Kilian Q Weinberger, and Yoav Artzi. Bertscore: Evaluating text generation with bert. *arXiv preprint arXiv:1904.09675*, 2019. 6
 - [51] Xiaoman Zhang, Chaoyi Wu, Ziheng Zhao, Weixiong Lin, Ya Zhang, Yanfeng Wang, and Weidi Xie. Pmc-vqa: Visual instruction tuning for medical visual question answering. *arXiv preprint arXiv:2305.10415*, 2023. 1, 2, 3
 - [52] Xiaoman Zhang, Chaoyi Wu, Ziheng Zhao, Jiayu Lei, Ya Zhang, Yanfeng Wang, and Weidi Xie. Radgenome-chest ct: A grounded vision-language dataset for chest ct analysis. *arXiv preprint arXiv:2404.16754*, 2024. 2, 3
 - [53] Xiangyu Zhao, Bo Liu, Qijiong Liu, Guangyuan Shi, and Xiao-Ming Wu. Easygen: Easing multimodal generation with bidiffuser and llms. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1351–1370, 2024. 1
 - [54] Zihao Zhao, Sheng Wang, Jinchun Gu, Yitao Zhu, Lanzhuju Mei, Zixu Zhuang, Zhiming Cui, Qian Wang, and Dinggang Shen. Chatcad+: Towards a universal and reliable interactive cad using llms. *IEEE Transactions on Medical Imaging*, 2024. 2
 - [55] Juexiao Zhou, Xiaonan He, Liyuan Sun, Jiannan Xu, Xiuying Chen, Yuetan Chu, Longxi Zhou, Xingyu Liao, Bin Zhang, and Xin Gao. Skingpt-4: an interactive dermatology diagnostic system with visual large language model. *arXiv preprint arXiv:2304.10691*, 2023. 3
 - [56] Deyao Zhu, Jun Chen, Xiaoqian Shen, Xiang Li, and Mohamed Elhoseiny. Minigpt-4: Enhancing vision-language understanding with advanced large language models. *arXiv preprint arXiv:2304.10592*, 2023. 1, 5, 6, 7
 - [57] Ke Zou, Yang Bai, Zhihao Chen, Yang Zhou, Yidi Chen, Kai Ren, Meng Wang, Xuedong Yuan, Xiaojing Shen, and Huazhu Fu. Medrg: Medical report grounding with multi-modal large language model. *arXiv preprint arXiv:2404.06798*, 2024. 3